



Perfect sound reflection

Researchers have demonstrated a 3D-printed metamaterial that can reflect and redirect sound almost perfectly. <https://digit.in/3dprintdev>



UFO spotted on the Moon

Scottish researcher, George Graham, claims to have seen a UFO-like object in the moon's crater in a NASA image. <https://digit.in/ufomoon>

REANIMATION

Bringing the extinct species of the planet back to life

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ecently, the news of the last male Northern White Rhinoceros passing away sent the world into a now-

routine tizzy about the need to ensure that animals aren't endangered or worse, driven to extinction. We've seen, time and again, how this kind of social media activity fizzles out in no time without any actual measures being taken. It is no secret that we've done a terrible job at conservation. If you still need statistics, a total of 801 species have gone extinct since the emergence of humans. Species are disappearing at a rate of 1,000 to 10,000 times more than the Earth's natural extinction rate, according to the World Wildlife Fund - certainly faster than what our conservation efforts can keep pace with. Which is why, a certain section of the scientific community is turning its attention towards de-extinction.

BRINGING BACK THE DEAD

De-extinction, also known as resurrection biology, sounds like something right out of a fantasy world's necromancy schools. Without putting a fantastic spin to it, it suffices to say de-extinction is the area of science that deals with bringing back species that have disappeared or similar species. Why do we say similar? To understand that, we need to understand the methods at play here. But first, should we bring extinct animals back?

The question has been asked time and again by concerned people



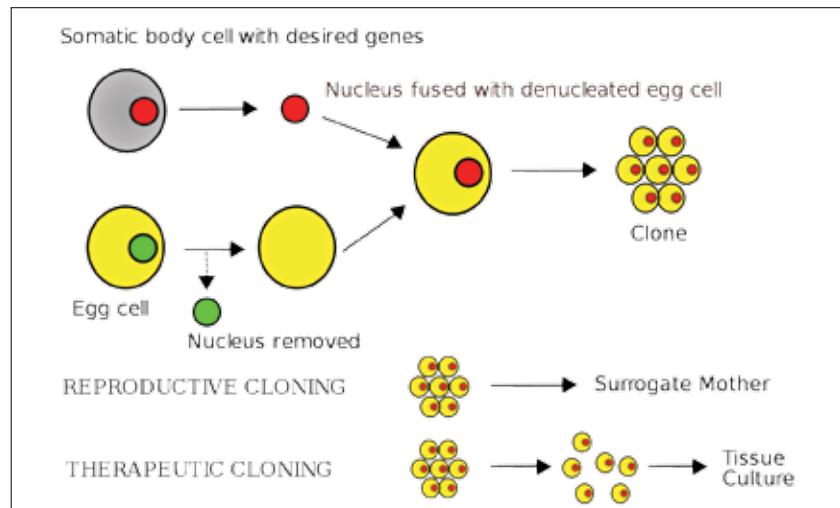
The Pyrenean ibex

of both scientific and non scientific inclinations - wouldn't bringing back extinct species interfere with nature? Wouldn't it be akin to playing God? (that typically does not work out so well). The answer is mostly in the negative. That is because we have already interfered too much with negative consequences. We caused the extinction in the first place and it is logically imperative that we get behind efforts to bring them back. Which brings us to the important question - how?

SOMATIC CELL NUCLEAR TRANSFER (SCNT)

Cloning is one of biology's most talked about areas. Cloning beings that are alive has already faced a copious amount of speculation on the ethics of the endeavor. You can imagine what the uninformed consensus about the same technology to dead species is. That hasn't deterred path breaking work in this area.

First things first - what we are going to talk about here isn't really cloning. It is mainly a Somatic Cell Nuclear Transfer (SCNT) where one set of cells is coming from the preserved remains of an extinct species. One of the recent examples of this process comes with the de-extinction of the bucardo. The bucardo was a wild goat, the Pyrenean ibex, that was driven to extinction by hunters in the mountain range that divides France from Spain. After Celia, the last living bucardo, passed away in an accident in the mountains, a team of Spanish and French scientists started working on bringing the species back from extinction. Led by José Folch, the team managed to inject the nuclei from Celia's preserved cells to goat eggs that had been emptied of their own DNA,



An oversimplified representation of SCNT